In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - I

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 02 and 09 to 30)** contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (03 to 08) contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 marks** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

(One Option Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

1. The relation between time t and distance x moved by a particle is $t = \alpha x^2 + \beta x$ where α and β are constants. The retardation is (if v represents velocity)
   (A) $2\alpha V^3$ (B) $2\beta V^3$
   (C) $2\alpha\beta V^3$ (D) $2\beta^2 V^3$

2. An object is cooled from 75^0C to 65^0C in 2 minutes in a room at 30^0C. The time taken to cool another identical object from 55^0C to 45^0C in the same room in minutes is
   (A) 4 (B) 5
   (C) 6 (D) 7

3. A source of sound is moving towards a stationary observer with a speed of 50 m/s. The observer measures the frequency of the source as 1000 Hz. The speed of sound is 350 m/s. The apparent frequency measured by the observer when the source is moving away after crossing the observer is
   (A) 750 Hz (B) 850 Hz
   (C) 1150 Hz (D) 1250 Hz

4. A spherical solid ball of volume V is made of a material of density ρ_1. It is falling through a liquid of density $\rho_2(\rho_2 < \rho_1)$. Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v. i.e., $F_{viscous} = -kv^2$ $(k > 0)$. The terminal speed of the ball is
   (A) $\sqrt{\frac{Vg(\rho_1 - \rho_2)}{k}}$ (B) $\frac{Vg\rho_1}{k}$
   (C) $\sqrt{\frac{Vg\rho_1}{k}}$ (D) $\frac{Vg(\rho_1 - \rho_2)}{k}$

Space for Rough work

5. A spring of force constant k is cut into two pieces such that one piece is double the length of the other. Then the long piece will have a force constant of

(A) $\frac{2}{3}k$ (B) $\frac{3}{2}k$

(C) 3k (D) 6k

6. A particle is projected horizontally from the top of a tower with a velocity v_0. If v be its velocity at any instant, then the radius of curvature of the path of the particle at the point (where the particle is at that instant) is directly proportional to

(A) v^3 (B) v^2

(C) v (D) $\frac{1}{v}$

7. One half mole of a monoatomic ideal gas absorbs 1200 J of heat energy while performing 2196 J of work. The temperature of the gas changes by

(A) 160^0C (B) 80^0C

(C) 40^0C (D) -160^0C

8. A train of mass m is moving on a circular track of radius R with constant speed v. The length of the train is half of the perimeter of the track. The linear momentum of the train will be

(A) Zero (B) mvR

(C) $2m\frac{v}{\pi}$ (D) mv

9. A piano string 1.5m long is made of steel of density 7.7×10^3 kg/m^3 and Young's modulus 2×10^{11}N/m^2. It is maintained at a tension which produces an elastic strain of 1% in the string. The fundamental frequency of transverse vibrations of string is

(A) 85Hz (B) 170Hz

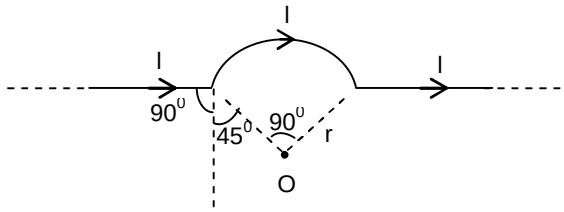
(C) 340Hz (D) 310Hz

Space for Rough work

10. Two organ pipes each closed at one end give 5 beats/sec when emitting their fundamental notes. If their length are in the ratio 50:51, their fundamental frequencies are
(A) 250,255 (B) 255,260
(C) 260,265 (D) 265,270

11. The escape velocity on earth is 11.2 km/s. What would be the escape velocity on a planet whose mass is 1000 times and radius is 10 times than that of earth
(A) 112 km/s (B) 11.2 km/s
(C) 1.12 km/s (D) 3.7 km/s

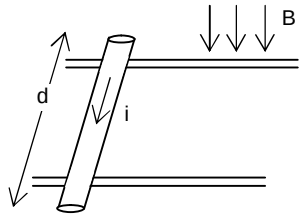
12. The magnetic field at the centre O of the arc in figure is (r is the radius of circular arc)



(A) $\frac{\mu_0 I}{4\pi \times r}\left[\sqrt{2}+\pi\right]$ (B) $\frac{\mu_0 I}{2\pi r}\left[\frac{\pi}{4}+1\left(\sqrt{2}-1\right)\right]$

(C) $\frac{\mu_0}{4\pi}\times\frac{I}{r}\left[\sqrt{2}+r\right]$ (D) $\frac{\mu_0}{4\pi}\times\frac{I}{r}\left[\sqrt{2}+\frac{\pi}{4}\right]$

13. A solid cylindrical uniform rod of mass 0.72 kg, length 50 cm and radius 6 cm rests on two parallel rails, as shown. The rod caries a current I = 48A (In the direction shown) and rolls along the rails without slipping. If it starts from rest due to applications of uniform magnetic field of magnitudes 0.25 T (directed perpendicular to the rod and the rail), then the friction force (In N) between rod and rails is

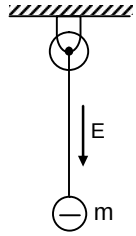


(A) 2 (B) 3
(C) 4 (D) 5

14. The critical angle for a prism and its surrounding interface is 36°. The minimum angle of prism for which no emergent ray is possible is
(A) 18° (B) 36°
(C) 72° (D) 144°

Space for Rough work

15. A simple pendulum charged negatively to q coulomb oscillates with a time period T in a downward electric field. If the electric field is with-drawn, the new time period is:



(A) = T (B) > T
(C) < T (D) none of these

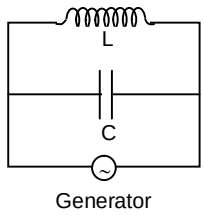
16. Two electrons are moving with the same speed V. One electron enters a region of uniform electric field while the other enters a region of uniform magnetic field. After sometime if the de Broglie wavelength of the two are λ_1 and λ_2, then (select the best alternative)

(A) $\lambda_1 = \lambda_2$ (B) $\lambda_1 > \lambda_2$
(C) $\lambda_1 > \lambda_2$ or $\lambda_1 < \lambda_2$ (D) $\lambda_1 > \lambda_2$ or $\lambda_1 < \lambda_2$ or $\lambda_1 = \lambda_2$

17. The efficiency of carnot engine is 0.6. It rejects 20 J of heat to the sink. The work done by the engine is

(A) 20 J (B) 30 J
(C) 33.5 J (D) 50 J

18. For the circuit shown in the figure, the current through the inductor is 0.6 A, while the current through the capacitor is 0.4 A. The current drawn from the generator is



(A) 1.0 A (B) 0.4 A
(C) 0.6 A (D) 0.2 A

19. A uniform solid brass sphere is rotating with angular speed ω_0 about a diameter. If its temperature is now increased by 100^0C. What will be its new angular speed (Given: $\alpha_{brass} = 2.0 \times 10^{-5} \text{per}^0\text{C}$)

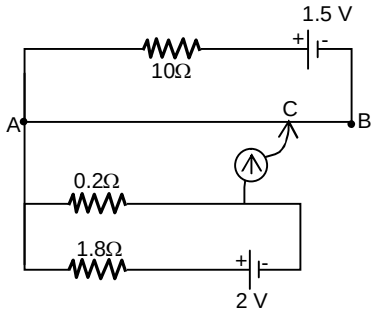
(A) 1.1 ω_0 (B) 1.01 ω_0
(C) 0.996 ω_0 (D) 0.824 ω_0

Space for Rough work

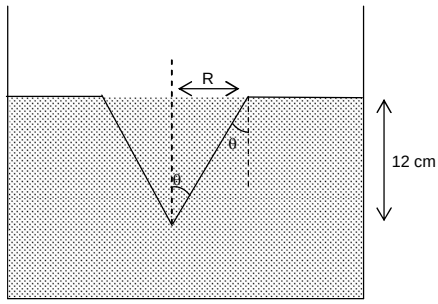
20. Which of the following phenomenon is not common to sound and light waves?
(A) interference (B) diffraction
(C) polarization (D) reflection

21. In a two slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the slits. If the screen is moved by 5×10^{-2} m towards the slits the change in fringe width is 3×10^{-5} m. If the distance between the slits is 10^{-3} m, the wavelength of light used is
(A) 4000 Å (B) 5550 Å
(C) 6000 Å (D) 6500 Å

22. In the figure, AB is a uniform wire of length 100 cm whose resistance is 5 Ω. If galvanometer reads zero, then AC =
(A) 10cm (B) 20cm
(C) 40cm (D) 80cm

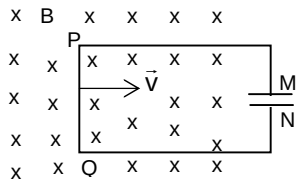


23. A fish looking up through the water sees the outside world contained in a circular horizon. If the refractive index of water is 4/3 and the fish is 12 cm below the surface, the radius of this circle in cm is
(A) $36\sqrt{5}$
(B) $4\sqrt{5}$
(C) $36\sqrt{7}$
(D) $36/\sqrt{7}$



Space for Rough work

24. A rod PQ is connected to the capacitor plates. The rod is placed in a magnetic field (B) directed downward perpendicular to the plane of the paper. If the rod is pulled out of magnetic field with velocity v as shown in figure.



(A) plate M will be positively charged
(B) plate N will be positively charged
(C) both plates will be similarly charged
(D) no charge will be collected on plates

25. A tank has a hole made at its bottom. The time needed to empty the tank from level h_1 to h_2 will be proportional to

(A) $h_1 + h_2$ (B) $\sqrt{h_1} + \sqrt{h_2}$

(C) $h_1 - h_2$ (D) $\sqrt{h_1} - \sqrt{h_2}$

26. In hydrogen like atoms the ratio of difference of energies $E_{4n} - E_{2n}$ and $E_{2n} - E_n$ varies with atomic number z and principle quantum number n as

(A) $\frac{z^2}{n^2}$ (B) $\frac{z^4}{n^4}$

(C) $\frac{z}{n}$ (D) none of these

27. A star initially has 10^{40} deutrons. It produces energy via the processes

$_1H^2 + {}_1H^2 \rightarrow {}_1H^3 + p$ and $_1H^2 + {}_1H^3 \rightarrow {}_2He^4 + n$.

If the average power radiated by the star is 10^{16} W, the deuteron supply of the star is exhausted in a time of the order of [The masses of nuclei are: $m(H^2) = 2.014$ u, $m(p) = 1.007$ u, $m(n) = 1.008$ u, $m(He^4) = 4.001$ u]

(A) 10^6 s (B) 10^8 s

(C) 10^{12} s (D) 10^{16} s

Space for Rough work

28. If the maximum kinetic energy of emitted photo electrons from a metal surface of work function 2.5 eV, is 1.7 eV. If wavelength of incident radiation is halved, then stopping potential will be
(A) 2.5 V (B) 5.9 V
(C) 5 V (D) 1.1 V

29. A light emitting diode (LED) has a voltage drop of 2V across it and passes a current of 10 mA. When it operates with a 6V battery through a limiting resistor R, the value of R is:
(A) 40 k Ω (B) 4 k Ω
(C) 200 Ω (D) 400 Ω

30. The number densities of electrons and holes in a pure germanium at room temperature are equal and its value is 3×10^{16} per m^3. On doping with aluminium, the hole density increases to 4.5×10^{22} per m^3. Then the electron density in doped germanium is
(A) $2.5 \times 10^{10}\ m^{-3}$ (B) $2 \times 10^{10}\ m^{-3}$
(C) $4.5 \times 10^{9}\ m^{-3}$ (D) $3 \times 10^{9}\ m^{-3}$

Space for Rough work

Chemistry

PART – II

SECTION – A

(One Option Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

1. 100 mL of $FeSO_4$ solution required 200 mL of 0.1 M acidified $KMnO_4$ solution for complete reaction. What is the molarity of the $FeSO_4$ solution?

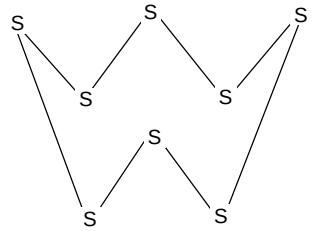
$$MnO_4^- + Fe^{2+} + H^+ \longrightarrow Mn^{2+} + Fe^{3+} + H_2O$$

(A) 0.5 M (B) 1 M
(C) 2 M (D) 1.5 M

2. Which of the following molecule has the highest value of dipole moment?
(A) PCl_3F_2 (B) PCl_2Br_3
(C) PCl_2F_3 (D) PCl_5

3. Which gas shows real behaviour?
(A) 16 g of dioxygen at STP occupies 11.2 L
(B) One gram dihydrogen in 0.5 L flask exerts pressure of 24.63 atm at 300 K
(C) One mole of ammonia at 300 K and 1 atm pressure occupies 22.4 L
(D) 5.6 L of CO_2 at STP is equal to 11 g

4. The oxidation number and hybridization of sulphur in S_8 molecule are respectively:



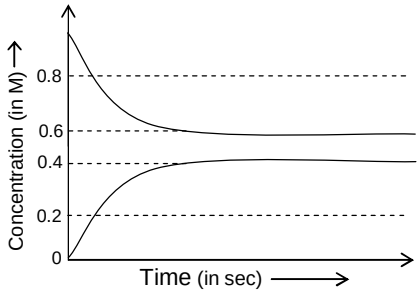
(A) +2 and sp^2 (B) +2 and sp^3
(C) zero and sp^2 (D) zero and sp^3

Space for Rough work

5. Which of the following is most soluble in water?
(A) $BeSO_4$ (B) $MgSO_4$
(C) $CaSO_4$ (D) $BaSO_4$

6. $X(g) + 2Y(g) \longrightarrow Products$
The rate equation of above reaction is given as follows:
Rate = $k[X][Y]^{0.5}$
What is the unit of rate constant of above reaction?
(A) $mol^{1/2}L^{-1/2}s^{-1}$ (B) $mol^{-1/2}L^{1/2}s^{-1}$
(C) $mol^{3/2}L^{-3/2}s^{-1}$ (D) $mol^{-3/2}L^{3/2}s^{-1}$

7. $HCN(aq) \rightleftharpoons H^+(aq) + CN^-(aq)$
The ionization constant (K_a) of above acid increases by
(A) adding more water (B) adding more HCN
(C) removing H^+ ions by adding bases (D) none of these

8. 

Above graph is given for the following reversible reaction.
$X(g) \rightleftharpoons Y(g)$
The equilibrium constant K_C of above reaction is:
(A) $\frac{1}{4}$ (B) $\frac{3}{2}$
(C) $\frac{2}{3}$ (D) 4

Space for Rough work

9. Which of the following relation is correct for an equilibrium system?
(A) $\Delta S_{system} = 0$ (B) $\Delta S_{surrounding} = 0$
(C) $\Delta S_{Total} = 0$ (D) $\Delta S_{system} = \Delta S_{surrounding}$

10. Which of the following compound on hydrolysis forms products that are polymerized to silicones?
(A) $SiCl_4$ (B) $(CH_3)_2SiCl_2$
(C) SiH_2Cl_2 (D) H_2SiF_6

11. $X(g) \rightleftharpoons Y(g) + Z(g)$
The equilibrium constant K_p of above reaction is 3 atm at a certain temperature. If equal moles of X, Y and Z are present at equilibrium, the equilibrium pressure would be:
(A) 12 atm (B) 3 atm
(C) 6 atm (D) 9 atm

12. $NaOH + CO \longrightarrow Product$
Which of the following is/are the product(s) of above reaction?
(A) HCOONa (B) $NaHCO_3$
(C) $Na_2CO_3 + H_2$ (D) $NaH + CO_2$

13. $SO_3^{2-} + H^+ \longrightarrow (X) + (Y)$
The unknown compounds of above reaction are:
(A) H_2S and O_2 (B) SO_2 and H_2O
(C) SO_3 and H_2 (D) SO_2 and H_2

14. For which of the following aqueous solution, there is maximum difference between the normal and abnormal colligative properties?
(A) 0.1 m $C_6H_{12}O_6$ (B) 0.1 m CH_3COOH
(C) 0.1 m HCOOH (D) 0.1 m CH_3OH

15. Which of the following aqueous solution is colourless?
(A) $CuSO_4$ (B) $Zn(NO_3)_2$
(C) $FeCl_2$ (D) $KMnO_4$

Space for Rough work

16. The relation between the molar conductance ($\wedge_m$) and equivalent conductance ($\wedge_e$) of the solution of $MgCl_2.KCl.6H_2O$ is
(A) $\wedge_m = \wedge_e \times 8$ (B) $\wedge_m = \wedge_e \times 3$
(C) $\wedge_e = \wedge_m \times 3$ (D) $\wedge_e = \wedge_m \times 8$

17. The strongest reducing agent out of the following is:
(A) NH_3 (B) PH_3
(C) AsH_3 (D) BiH_3

18. Which of the following interhalogen compound does NOT exist?
(A) IF_3 (B) IF_4
(C) IF_5 (D) IF_7

19. The hybridization of xenon in XeO_3 is:
(A) sp^3d^2 (B) sp^3d
(C) sp^3d^3 (D) none of these

20. $Pb(NO_3)_2\,(\text{excess}) + CrO_2Cl_2 + NaOH \xrightarrow{25^oC} \text{Products}$
Which of the following precipitates are formed in the above reaction?
(A) $PbCrO_3$ and $PbCl_2$ (B) $PbCrO_4$ and $PbCl_2$
(C) $PbCr_2O_7$ and $PbCl_4$ (D) Na_2PbO_2 and $PbCl_4$

21. How many effective no. of Na^+ ions are present in the unit cell of NaCl if the ions along all the body diagonals of the unit cell are removed?
(A) 4 (B) 3
(C) 2 (D) 1

22. The fraction of chlorine precipitated by $AgNO_3$ solution from $[Cr(NH_3)_5Cl]Cl_2$ is
(A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{2}{3}$ (D) $\frac{1}{4}$

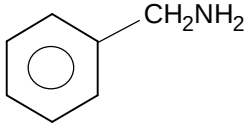
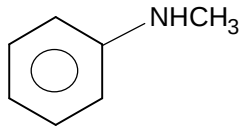
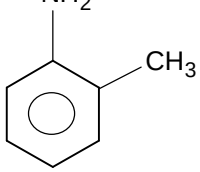
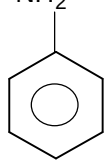
Space for Rough work

23. $Zn(s)|\ Zn^{2+}(1M)||\ Cu^{2+}(1M)|Cu(s)$
$E^0_{Zn^{2+}/Zn} = -0.76\,V$ and $E^0_{Cu^{2+}/Cu} = 0.34\,V$
Which of the following relation(s) is/are correct for the above electrochemical cell?
(A) $E^0_{cell} > 0$ (B) $E_{cell} > 0$
(C) $E_{cell} = E^0_{cell}$ (D) All are correct

24. Which one of the following has aromatic character ?
(A) Cyclopentadienyl cation (B) Cyclopentadienyl radical
(C) Cyclopentadienyl anion (D) Cyclopentadiene

25. How many optical isomer(s) is/are possible for fructose?
(A) 10 (B) 8
(C) 16 (D) 12

26. Number of C = C bonds present in anthracene is
(A) 5 (B) 3
(C) 9 (D) 7

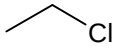
27. Which of the following is the strongest base?
(A) 
(B) 
(C) 
(D) 

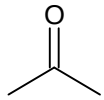
Space for Rough work

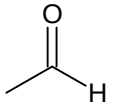
28. In order to response to xanthoproteic test. the amino acids should contain
(A) aliphatic vinyl group (B) phenyl group
(C) any cyclic group (D) allyclic group

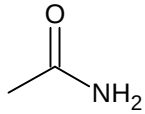
29. $[X] \xrightarrow[(ii)\ H^{+}(aq)]{(i)\ NH_4Cl;KCN(equimolar)} CH_3CH(NH_2)COOH$

Compound[X] is

(A) 

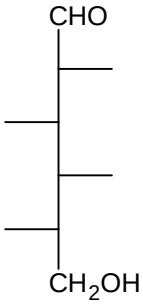
(B) 

(C) 

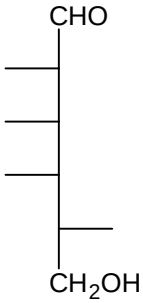
(D) 

Space for Rough work

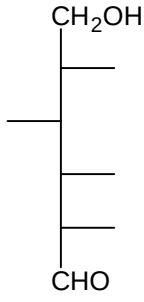
30. Compound which gives same osazone as given by



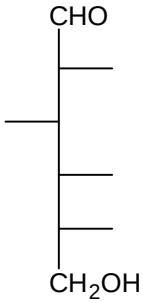
(A)



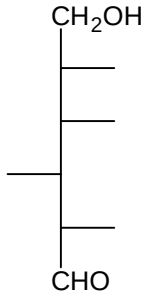
(B)



(C)



(D)



Space for Rough work

Mathematics

PART – III

SECTION – A

(One Option Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

1. Find the sum of the series $\cot^{-1}\frac{31}{12}+\cot^{-1}\frac{139}{12}+\cot^{-1}\frac{319}{12}+.....+\cot^{-1}\left(3n^2-\frac{5}{12}\right)$.

   (A) $\cot^{-1}\frac{13n+18}{12n}$ (B) $\cot^{-1}\frac{18n+13}{12n}$

   (C) $\cot^{-1}\frac{36n-5}{12n}$ (D) $\cot^{-1}\frac{28n+3}{12n}$

2. A curve satisfying the initial condition, $y(1)=0$, satisfies the differential equation, $x\frac{dy}{dx}=y-x^2$. The area bounded by the curve and the x – axis is:

   (A) $\frac{1}{2}$ (B) $\frac{1}{3}$

   (C) $\frac{1}{4}$ (D) $\frac{1}{6}$

3. The function f (x) = $\frac{\log(\pi+x)}{\log(e+x)}$ $(x \geq 0)$ is

   (A) increasing in $[0, \infty)$

   (B) decreasing in $[0, \infty)$

   (C) increasing in $\left[0, \frac{\pi}{e}\right)$ and decreasing in $\left[\frac{\pi}{e}, \infty\right)$

   (D) decreasing in $\left[0, \frac{\pi}{e}\right)$ and increasing in $\left[\frac{\pi}{e}, \infty\right)$

Space for Rough work

4. The total number of solutions of $|\cot x| = \cot x + \frac{1}{\sin x}$, $x \in [0, 3\pi]$ is

(A) 0 (B) 1
(C) 2 (D) 3

5. Let $a \in (0, 1]$ satisfies the equation $a^{2016} - 2a + 1 = 0$ and $S = 1 + a + a^2 + ... + a^{2015}$ Sum of all possible value(s) of S, is

(A) 2015 (B) 2016
(C) 2017 (D) 2018

6. Locus of all point P (x, y) satisfying $x^3 + y^3 + 3xy = 1$ consists of union of

(A) a line and an isolated point (B) a line pair and an isolated point
(C) a line and a circle (D) a circle and an isolated point

7. If $f(x) = x^2 + bx + c$ and $f(2 + t) = f(2 - t)$ for all real numbers t, then which of the following is true?

(A) $f(1) < f(2) < f(4)$ (B) $f(2) < f(1) < f(4)$
(C) $f(2) < f(4) < f(1)$ (D) $f(4) < f(2) < f(1)$

8. The average sales and standard deviation of sales for four months for a company are as follows :

	Month 1	Month 2	Month 3	Month 4
Average Sales :	30	57	82	28
Standard deviation of sales:	2	3	4	2

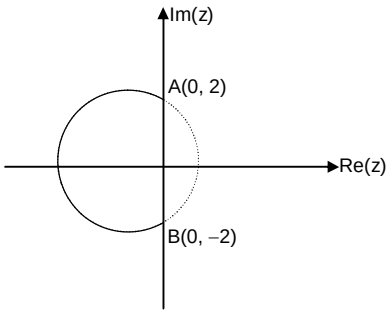
During which month are the sales most consistent?

(A) month 1 (B) month 2
(C) month 3 (D) month 4

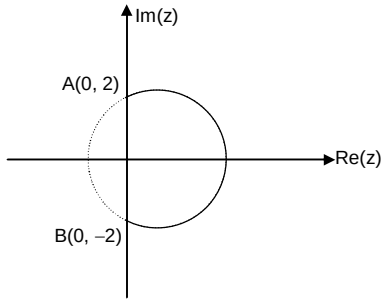
Space for Rough work

9. If $\arg\left(\frac{z-2i}{z+2i}\right)=\frac{\pi}{6}$, then which of the following options gives the correct locus of z ?

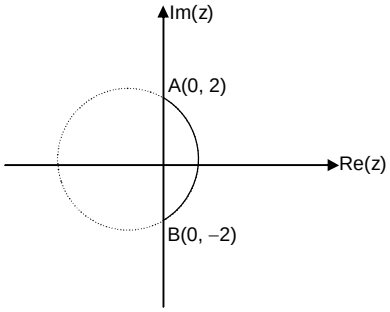
(A)



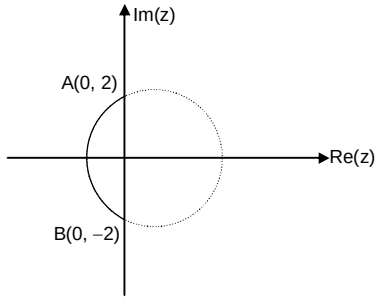
(B)



(C)



(D)



10. Let $S=\frac{1}{\sin 8^\circ}+\frac{1}{\sin 16^\circ}+\frac{1}{\sin 32^\circ}+.....+\frac{1}{\sin 4096^\circ}+\frac{1}{\sin 8192^\circ}$.

If $S=\frac{1}{\sin\alpha}$ where $\alpha \in (0, 90^\circ)$, then the value of α (in degrees) is

(A) 4° (B) 2°

(C) 6° (D) None of these

Space for Rough work

11. If $f(x)$ is a differentiable function such that $f'(2)=6$ and $f'(1)=4$, then $\lim\limits_{h\to 0}\dfrac{f\left(h^3+3h+2\right)-f(2)}{f\left(2h-2h^2+1\right)-f(1)}$ equals:

(A) $\dfrac{4}{3}$ (B) $\dfrac{4}{9}$

(C) $\dfrac{3}{4}$ (D) $\dfrac{9}{4}$

12. If $I=\int \sin x.\cos x.\cos 2x.\cos 4x.\cos 8x.\cos 16x\,dx$, then the value of I is

(A) $\dfrac{\sin 16x}{1024}+c$ (B) $-\dfrac{\cos 32x}{1024}+c$

(C) $\dfrac{\cos 32x}{1096}+c$ (D) $-\dfrac{\cos 32x}{1096}+c$

13. If the equation $x^4+8x^3+18x^2+8x+a=0$ has four distinct real roots, then the range of a is:

(A) (0, 9) (B) (–9, 0)

(C) (–8, 1) (D) (–1, 8)

14. $y=f(x)$ is a function which satisfies:

(i) $f(0)=0$

(ii) $f''(x)=f'(x)$ and

(iii) $f'(0)=1$

then the area bounded by the graph of $y=f(x)$, the lines $x=0$, $x-1=0$ and $y+1=0$, is:

(A) e (B) $e-2$

(C) $e-1$ (D) $e+1$

Space for Rough work

15. Three parallel chords of a circle have lengths 2,3,4 and subtend angles $\alpha, \beta, \alpha+\beta$ at the centre respectively (given $\alpha+\beta<\pi$), then $\cos\alpha$ is equal to

(A) $\frac{15}{31}$ (B) $\frac{17}{35}$

(C) $\frac{17}{32}$ (D) none of these

16. Let z_1, z_2, z_3 be three distinct points $|z|=1$. If θ_1, θ_2 and θ_3 be the arguments of z_1, z_2, z_3 respectively then $\cos(\theta_1-\theta_2)+\cos(\theta_2-\theta_3)+\cos(\theta_3-\theta_1)$

(A) $\geq -\frac{3}{2}$ (B) $\leq -\frac{3}{2}$

(C) $\geq \frac{3}{2}$ (D) ≤ 2

17. Number of solutions of the equation

$\log_{10}\left(\sqrt{5\cos^{-1}x-1}\right)+\frac{1}{2}\log_{10}\left(2\cos^{-1}x+3\right)+\log_{10}\sqrt{5}=1$ is:

(A) 0 (B) 1

(C) more than one but finite (D) infinite

18. If the function $f(x)=-4e^{\frac{1-x}{2}}+1+x+\frac{x^2}{2}+\frac{x^3}{3}$ and $g(x)=f^{-1}(x)$, then the value of $g'\left(-\frac{7}{6}\right)$ equals

(A) $\frac{1}{5}$ (B) $-\frac{1}{5}$

(C) $\frac{6}{7}$ (D) $-\frac{6}{7}$

Space for Rough work

19. If z is a complex number, then $\frac{z\left(|z|^2-1\right)+2\mathrm{Re}(z)}{|z|^2}$ is equal to

(A) $z+\frac{1}{z}$ (B) $z-\frac{1}{z}$

(C) $\mathrm{Re}(z)$ (D) none

20. Two teachers are interviewing top 6 students in FTRE 2016 exam in two different subjects, starting at the same time. Each teacher interviews a student for 15 minutes. The number of ways in which the interviews can be scheduled is

(A) 6! (B) 5!

(C) 44 x 6! (D) 265 x 6!

21. The length of the perpendicular from the origin to the plane passing through the point $\vec{a}$ and containing the line $\vec{r}=\vec{b}+\lambda\vec{c}$ is

(A) $\frac{\left[\vec{a}\ \vec{b}\ \vec{c}\right]}{\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|}$ (B) $\frac{\left[\vec{a}\ \vec{b}\ \vec{c}\right]}{\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}\right|}$

(C) $\frac{\left[\vec{a}\ \vec{b}\ \vec{c}\right]}{\left|\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|}$ (D) $\frac{\left[\vec{a}\ \vec{b}\ \vec{c}\right]}{\left|\vec{c}\times\vec{a}+\vec{a}\times\vec{b}\right|}$

22. If $A=\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$, then $\lim_{n\to\infty}\frac{A^n}{n}$ is (where $\theta\in R$)

(A) a zero matrix (B) an identity matrix

(C) $\begin{bmatrix}0 & 1\\ 0 & -1\end{bmatrix}$ (D) $\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}$

Space for Rough work

23. Let $T = \int_0^{\ln 2} \frac{2e^{3x} + e^{2x} - 1}{e^{3x} + e^{2x} - e^{x} + 1} dx$, then e^T equals:

(A) $\frac{7}{4}$ (B) $\frac{7}{2}$

(C) $\frac{11}{2}$ (D) $\frac{11}{4}$

24. A line is drawn from the point P (1, 1, 1) and perpendicular to a line with direction ratios 1, 1, 1 to intersect the plane $x + 2y + 3z = 4$ at Q. The locus of Q is

(A) $\frac{x}{1} = \frac{y-5}{-2} = \frac{z+2}{1}$ (B) $\frac{x}{-2} = \frac{y-5}{1} = \frac{z+2}{1}$

(C) $x = y = z$ (D) $\frac{x}{2} = \frac{y}{3} = \frac{2}{5}$

25. The lines $2x^2 - 3y^2 + xy - 3x - 7y + 2 = 0$ represent two normals of a circle which passes through the focus of the parabola $(5x + 15)^2 + \left(5y - \frac{5}{2}\right)^2 = \left(3x + 4y - \frac{1}{2}\right)^2$. The equation of the circle is

(A) $4x^2 + 4y^2 - 8x - 8y - 65 = 0$ (B) $4x^2 + 4y^2 - 8x + 8y - 65 = 0$

(C) $x^2 + y^2 - 2x - 2y - 73 = 0$ (D) $x^2 + y^2 - 2x + y + 73 = 0$

26. The asymptotes of the hyperbola $y = x + \frac{1}{x}$ are

(A) $y = 0, y = x$ (B) $y = 0, y = -x$

(C) $x = 0, y = -x$ (D) $x = 0, y = x$

Space for Rough work

27. If $x\frac{dy}{dx} = y\left(\log_e y - \log_e x + 1\right)$, then the solution of the equation is

(A) $y\log_e\left(\frac{x}{y}\right) = cx$ (B) $x\log_e\left(\frac{y}{x}\right) = cy$

(C) $\log_e\left(\frac{y}{x}\right) = c\,x$ (D) $\log_e\left(\frac{x}{y}\right) = cy$

28. A man is dealt a poker hand (consisting of 5 cards) from an ordinary pack of 52 playing cards. The number of ways in which he can be dealt a "straight" (a straight is five consecutive values NOT of the same suit. e.g. {Ace 2, 3, 4, 5}, {2, 3, 4, 5, 6}……..and {10 . J. Q. K. Ace}) is

(A) $10\left(4^5 - 4\right)$ (B) $4!.2^{10}$

(C) 10.2^{10} (D) none of these

29. The minimum value of $\left(\tan A - \cos B\right)^2 + \left(\cot A - \sin B\right)^2$ for all permissible independent variables A and B is

(A) 0 (B) $\sqrt{2} - 1$

(C) $3 - 2\sqrt{2}$ (D) cannot be determined

30. From a set of numbers $\{1,2,3,.....,999\}$, six distinct numbers are chosen. Out of these six numbers three are chosen and labeled as a_1, a_2, a_3. The remaining are labeled as b_1, b_2, b_3. The probability that a brick of dimensions $a_1 \times a_2 \times a_3$ fits in a box of dimensions $b_1 \times b_2 \times b_3$ is equal to (assume that the brick can be rotated before putting into the box)

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{1}{5}$

Space for Rough work